PAN-C — User Manual

TEQSAS PRODUCTS

PAN-C

User Manual

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1 Before you start

Intended use

Only ever use the appliance as described in the operating instructions. Any other use is not in accordance with the intended use. TEQSAS accepts no liability for improper use or misuse of the product. Observe the respective country-specific regulations before use.

The device is designed as an expansion panel for Coda Audio Linus 12C power amplifiers. It has analog and digital signal inputs as well as network connections and outputs for loudspeaker signals. The PAN-C is supplied with power via a mains connection with Neutrik powerCON TRUE1 plug and has a loop-through for the power supply of the Coda Linus 12C power amplifier.

What do the symbols used mean?

Hazard warnings and notes are clearly marked in the instructions for use. The following symbols are used:

DANGER

Immediate danger to life or injury.

Immediately dangerous situation that will result in death or serious injury.

WARNING

Risk of death or injury!

The mains cable fed through from the front must not be crushed. Ensure that the cable is routed freely in the 19” rack. There is a recess in the housing of the PAN-C for this purpose.

CAUTION

Risk of damage to the appliance!

Situation that may result in material damage.

ATTENTION

Potential risk of injury!

Dangerous situation that may result in injury.

ADVICE

Information provided for a better understanding of the processes

2 For Your Safety

WARNING!

```
**Failure to observe the safety information and instructions may result in
electric shock, fire and/or serious injury!**

Read all safety information and instructions.
```

General safety instructions

- Keep all safety information and instructions for future reference. The term “appliance” used in the safety instructions refers to mains-powered appliances (with mains cable) and battery-powered appliances (without mains cable).
- Danger to life and accidents for infants and children! Never leave children unattended with packaging material and product. There is a risk of suffocation from packaging material and a risk to life from strangulation. Children often underestimate the dangers. Always keep children away from the product. The product is not a toy.
- To ensure safe handling of this appliance, the user must have read and understood these instructions for use before using the appliance for the first time.
- If you sell or pass on the appliance, it is essential that you also hand over these instructions for use.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and/or knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- The appliance may only be used if it is in perfect working order and fully assembled. If the appliance or any part of it is defective, it must be taken out of service and repaired or disposed of properly.
- Use the appliance only for the purpose for which it is intended.
- Keep children away from the appliance! Keep the device safe from children and unauthorised persons.
- Only use and store the device within the permissible ambient conditions (temperatures, humidity, etc.).

Workplace safety * Keep your work area clean and well lit. Untidiness or unlit work areas can lead to accidents. Observe the workplace and accident prevention regulations applicable in your country. * Do not work with the electrical appliance in potentially explosive atmospheres containing flammable liquids, gases or dust. Power tools generate sparks that can ignite the dust or vapours.

Electrical safety

- Static electricity can occur in dry environments. In dry rooms, touch a metal object to discharge the static electricity before operating the appliance.
- Do not misuse the cable to carry or hang up the appliance or to pull the plug out of the socket. Keep cables away from heat, oil, sharp edges or moving parts of the appliance. Damaged or tangled cables increase the risk of damage to the appliance.
- If you use the appliance outdoors, only use extension cables that are also suitable for outdoor use. The use of an extension cable suitable for outdoor use reduces the risk of damage to the appliance.

Safety of persons

- Be attentive, pay attention to what you are doing and use common sense when working. Do not use the appliance when you are tired or under the influence of drugs, alcohol or medication. A moment of inattention during use can lead to serious injury (e.g. from electric shock).

Service

- Only have your appliance repaired by qualified specialists and only with original spare parts. This will ensure that the safety of the appliance is maintained.
- Ensure adequate lighting when working with the appliance. Poor visibility can lead to an increased risk of accidents.
- Do not use any accessories that have not been specially designed and recommended by the manufacturer for this appliance. Just because you can connect the accessories to your appliance does not guarantee trouble-free and safe use.
- The device may show signs of wear and tear over time.
- The use of unauthorised accessories may cause damage or wear that is not covered by the warranty

> CAUTION!

```
**Risk of injury!**

Improper repairs can result in your appliance no longer functioning safely.
You endanger yourself and your surroundings
```

!!! abstract ""Attention!"" Risk of damage to the appliance!

```
All components are maintenance-free. Never open the appliance, otherwise the
warranty will be invalidated.
```

Special device instructions * Do not use the appliance in potentially explosive atmospheres or in the vicinity of flammable liquids or gases! * If you have an allergic reaction to the colour or metallic parts of the appliance, it may cause itching, eczema or swelling of the skin. If this is the case, stop using the appliance and consult a doctor. * Do not drop the device and do not subject it to shocks. * The device or parts of it may be damaged if dropped, bent or deformed. * Symbols on your device must not be removed or covered. Labels on the appliance that are no longer legible must be replaced immediately.

3 Product description

3.1 Scope of delivery

- TEQSAS PAN-C
- powerCON®TRUE1 cable Power cable
- powerCON® TRUE1 16A True1-powerCON® TRUE1 32A connector
- Instructions for use

3.2 Your device at a glance

The TEQSAS PAN-C v2, hereinafter referred to as PAN-C, facilitates the use of Coda Audio Linus 12C system amplifiers. The audio inputs and the loudspeaker signal are accessible on the front of the power amplifier. The option of choosing between analogue audio or AES/EBU (AES3) as the input signal means that many inputs and outputs can be accommodated in a 1U device height.

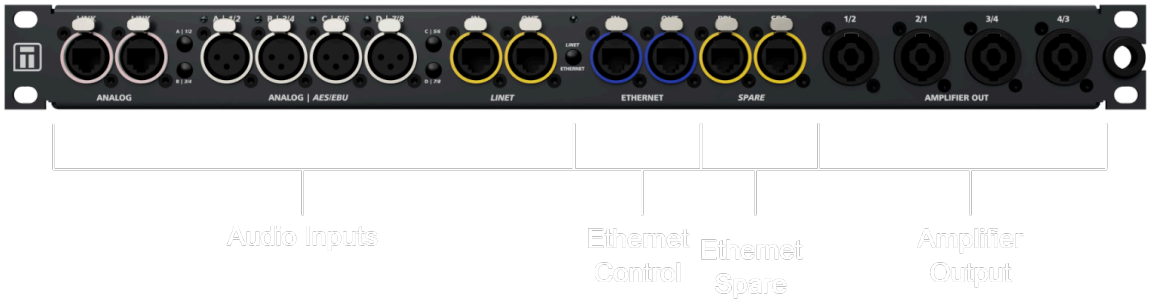


Figure 1: Figure 1 – Front View with Inputs und Outputs

Figure 1 – Front view with inputs und outputs

Figure 2: Figure 2 - Detailed front view with Buttons

Figure 2 - Detailed front view with buttons

Figure 3: Figure 3 - Back view

Figure 3 - Back view

- Front panel connections:
  - Audio Inputs
    - Analogue - XLR or RJ45
    - AES/EBU
    - LiNET
    - Network-based audio (if supported by the amplifier)
  - Ethernet
    - Connection of the amplifier to the network for Linus Control
    - LiNET-C
  - Save
    - Free network connection, e.g. for amplifiers with Dante or other network audio solutions.
  - Amplifier Out
    - Output of the loudspeaker channels, each fully assigned, four-pin Neutrik speakON®.
- Rear panel connections:
  - Power IN/OUT
    - Neutrik powerCON®TRUE1 16A Input
    - Neutrik powerCON®TRUE1 16A Output
  - Amplified Audio
    - Speaker signal from the amplifier to the front panel as four-pin Neutrik speakON®.
  - Spare
    - See front panel
  - Ethernet
    - See front panel
  - Audio Outputs
    - See front panel
- Buttons on the front:
  - Analogue | AES/EBU

    - Selection of analogue or AES/EBU at the XLR inputs
  - LINET / Ethernet
    - Choice between LiNET (8ch.) via the LiNET sockets or LiNET-C (4ch. + Control) via the Ethernet sockets

Technical data

TEQSAS PAN-C	
Permissible ambient temperature for use	0 - 40°C
Permissible ambient temperature for storage	–20 - 50°C
Dimensions	482 × 180 × 44mm
Weight PAN-C	3040gr

The manufacturer reserves the right to make technical changes to the product / product group without prior notice.

4 Setting it up

4.1 Installation

HINWEIS

Always use all four screws to secure the PAN-C in the 19” rack.

WARNING

Risk of death or injury!

The mains cable fed through from the front must not be crushed. Ensure that the cable is routed freely in the 19” rack. There is a recess in the housing of the PAN-C for this purpose.

The PAN-C is designed in 19-inch/1U format and is intended for installation in a Coda Audio LINUS M-RACK, but can be installed in any standard 19” rack.

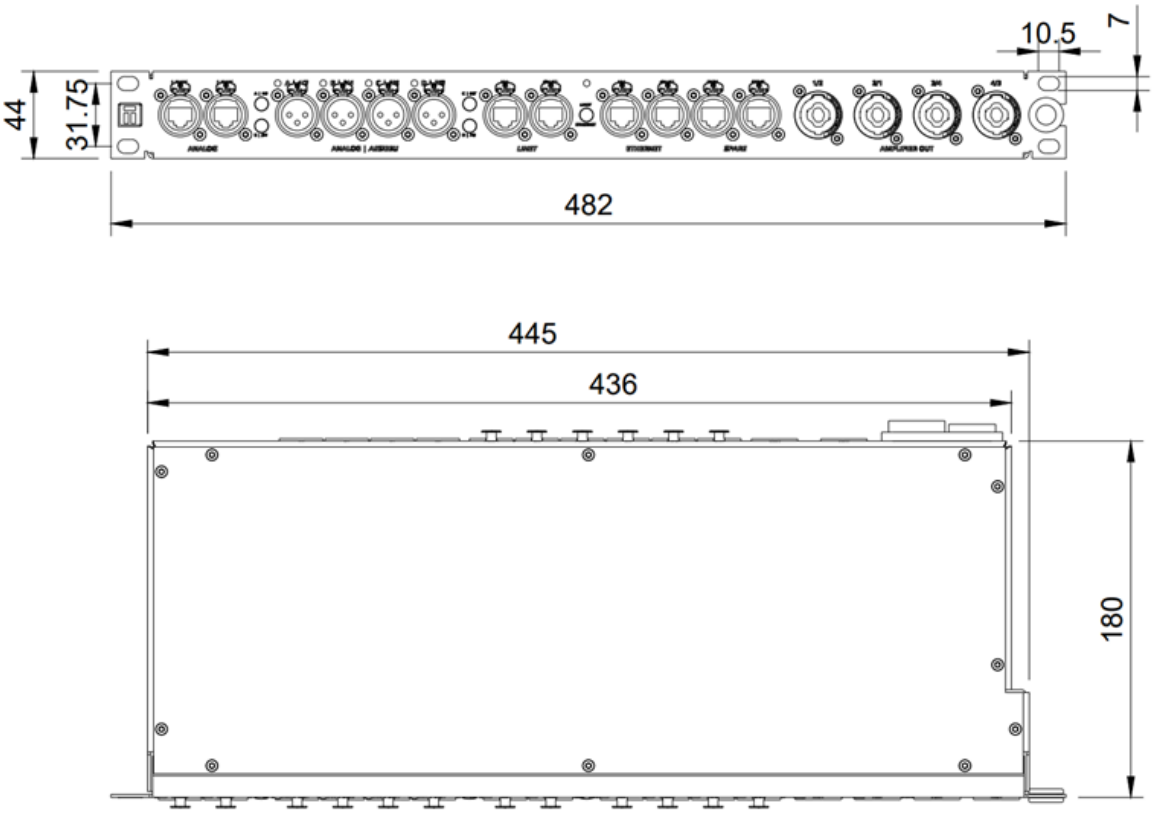


Figure 4: PAN-C Dimensions

Figure 4 - Dimensions in mm

4.2 Power supply

WARNING

Danger of electric shock!

The appliance complies with appliance protection class I and may only be operated in supply networks with protective earth. The mains cable or the mains plug must not be damaged. If in doubt, replace it with a faultless mains cable. Disconnect the appliance from the mains during thunderstorms or during longer periods when the appliance is not in use.

ATTENTION!

Risk of damage to the appliance!

Only connect the appliance to supply networks that correspond to the specified mains voltage range. This can be found on the information plate.

Mains voltage range: 230V~, 50Hz, 70mA.

The device connection is designed as a Neutrik powerCON® TRUE1, which is a switchable connector. However, this should not be disconnected during audio playback as Coda Audio explicitly points this out. Always switch off the LINUS 12C first using the switch on the front panel or the shutdown function provided for this purpose in the Linus Control.

4.3 Connection

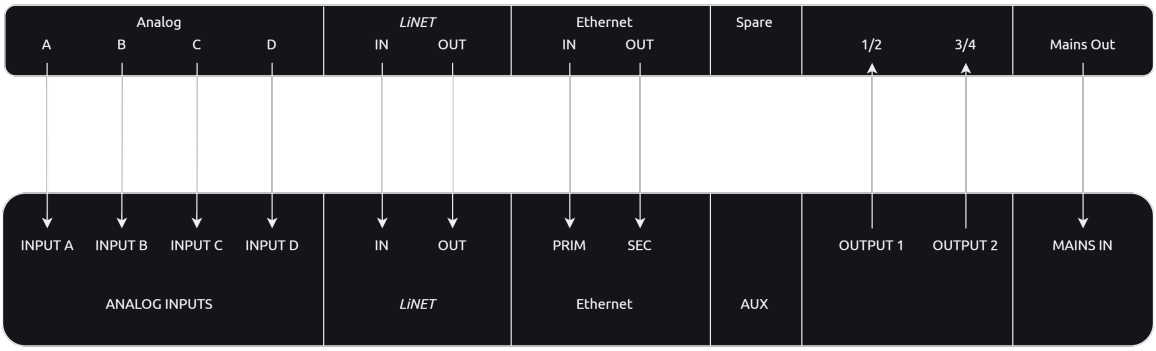
The PAN-C is intended to be mounted in the Coda M rack as a permanent replacement for the Coda PAN-M in combination with Linus 12C power amplifiers. For this purpose, all existing connections on the rear of the PAN-C can be wired according to the following diagram.

WARNING!

Danger of electric shock!

Only use high-quality cables with correctly connected plugs to connect the loudspeaker signal.

PAN-C Back



Linus 12C Back

5 Operation

The PAN-C is supplied with power as soon as you plug in the mains plug of the device. The power supply is only required for switching to the selected inputs. You can also use the PAN-C without power. In this case, however, only the analogue inputs and the LiNET-C are available.

5.1 Audio inputs

Analogue audio signals can be fed in either via the 4 XLR sockets or via the RJ45 sockets. The sockets are connected in parallel, so they can be used as a loop-through. Please ensure that one input channel is recorded via either RJ45 or XLR.

Digital audio signals can either be fed into the XLR sockets in AES3 (AES/EBU) format or as Coda Audio LiNET. Here too, as the Coda Audio power amplifier only has LiNET as a digital audio input, either AES3 or LiNET must be selected as the input signal. To use LiNET-C (4ch. + Control), the LiNET cable must be plugged into the Ethernet IN socket and the To transfer signals in AES3 format, use the buttons (see figure 2) on the respective XLR sockets to set them to either analogue input or AES3 input. The status LEDs above the XLR sockets indicate which mode the socket is in:

- White - Analogue Input
- Yellow - AES3 Input

> NOTE
>
> - In the de-energised state, the XLR inputs are always routed to the analogue inputs of the amp.
> - When de-energised, the LiNET mode is always set to LiNET-C.

Ethernet/Control can either be connected via a separate Ethernet connection or via LiNET-C. In both cases, connect the cables to the two Ethernet ports.

Use the LiNET/Ethernet button to switch the function of the Ethernet ports (see Figure 2). The colour of the status LED above the button indicates the mode of the Ethernet ports: * Yellow - LiNET, 8ch. LiNET via the LiNET ports, Control via Ethernet * Blue - LiNET-C, 4ch. LiNET and control via the Ethernet ports

> NOTE
>
> - The PAN-C has an EEPROM in which the last mode used is saved. This applies to both the Analogue/AES3 and LiNET/Ethernet selections. The device will start with the same input configuration the next time it is started

Spare are two looped Cat5e cables that have no direct application with the Coda Audio Linus 12c. However, these cables can be used freely, for example to run additional Ethernet cables to the rear of the rack.

5.2 Audio outputs

The four loudspeaker channels are routed to the front of the PAN-C in full. The sockets are designed as NL4. Each channel pair is also available as a rotated version.

> WARNING!
>
> Danger of electric shock!
>
> Dangerous voltages can occur at the output channels of amplifiers. Therefore, only use intact cables with correctly connected plugs. The output channels of the amplifier must not be short-circuited under any circumstances.

The four loudspeaker channels are routed to the front of the PAN-C fully loaded. The sockets are designed as NL4. Each channel pair is also available as a turned version.

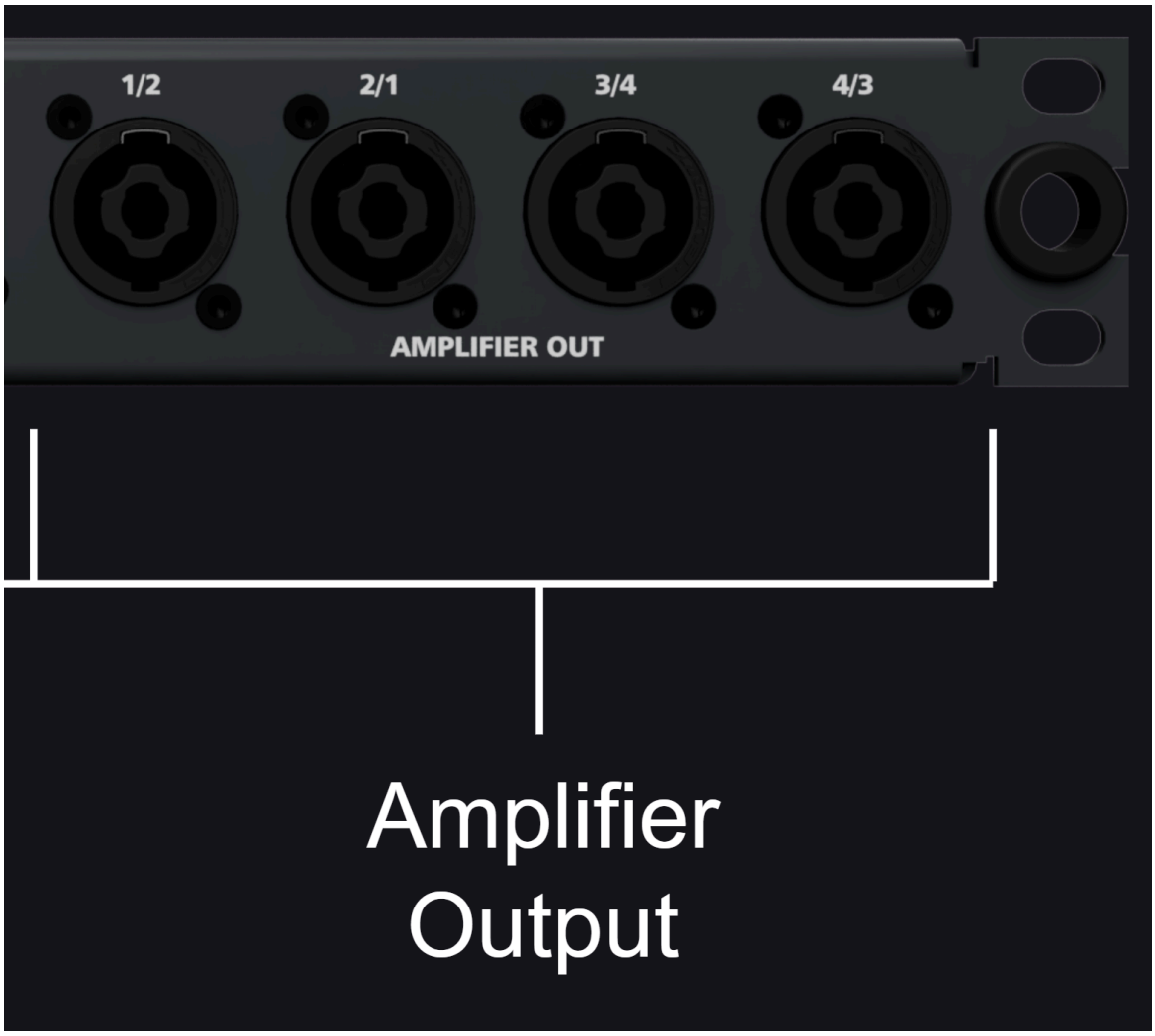


Figure 5: Amplifier Outputs

Cleaning and care

!!!danger “Danger!” Danger to life due to electric shock!

```
Disconnect the mains cable before carrying out any work on the appliance. Make
sure that the appliance is de-energised. This is the case when all LEDs on the
front panel are off.
```

!!!abstract “Attention!” Risk of damage to the appliance!

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Do not immerse the appliance in water when cleaning, as water can penetrate
the electrical system and damage it.
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!!!abstract “Attention!” Possible product damage!

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The surface can be damaged.
Do not use scratching or abrasive objects or aggressive cleaning agents for
cleaning.

* Use a soft, lint-free, dry cleaning cloth to clean the housing.
* Do not use chemicals, cleaning agents or solvents. This can lead to
discolouration and corrosion of the outside of the appliance or result in an
electric shock or fire.
```

Malfunctions and help

!!!warning “Caution!” Risk of injury! Improper repairs can result in your appliance no longer functioning safely. You are putting yourself and your surroundings at risk.

It is often only minor faults that lead to a malfunction. In most cases, you can easily rectify these yourself. Please refer to the following table before contacting your sales partner or dealer. This will save you a lot of trouble and possibly also costs.

If the appliance requires servicing, take it to your sales partner.

Error/Fault	Cause	Remedy
Device does not work	Device defective?	
Analogue audio signal does not reach the amplifier	Channel mode selected incorrectly	Switch corresponding channel from AES/EBU to analogue. Status LED must light up white
Analogue audio signal does not reach the amplifier	Amplifier input not set to analogue input	Set the corresponding channel to analogue in the control software or on the control panel.
AES/EBU signal does not reach the amplifier	See above	Set channel mode correctly, LEDs must light up yellow. Set the input on the amplifier correctly.

If you cannot rectify the fault yourself, please contact our support team directly. Please note that improper repairs will also invalidate the warranty and you may incur additional costs.

Support

Mail: support@teqsas.de

Please describe your fault as precisely as possible so that we can carry out troubleshooting. Please also provide us with the serial number and a contact where we can best reach you. ## Service centre

In the event of damage, please contact our service centre directly.

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TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth
Deutschland
Tel.: +49 (0)2233 611-500
E-Mail: service@teqsas.de
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Observe ambient conditions

The device must not be used in potentially explosive atmospheres containing flammable liquids, gases or dusts.

Storage

Clean and dry all parts and store them in a clean and dry place.

Disposal

The packaging is made of environmentally friendly materials that can be disposed of at local recycling centres. This appliance must not be disposed of with household waste. You are obliged to dispose of waste electrical and electronic equipment separately. You can find out how to dispose of old appliances from your local authority. By disposing of old appliances separately, you are sending them for recycling or other forms of reuse. In doing so, you will help to prevent harmful substances from being released into the environment.

EU Declaration of Conformity

Figure 6: CE

in accordance with EU Directive 2014/30/EU (Electromagnetic Compatibility) of 26 February 2014 and 2014/35/EU (Low Voltage Directive) of 26 February 2014.

We hereby declare that the device designated below, in its design and construction and in the version placed on the market by us, complies with the essential health and safety requirements of EU Directive 2014/30/EU and the requirements for electromagnetic compatibility. This declaration shall lose its validity if the device is modified without our agreement.

Manufacturer

TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth

The manufacturer bears sole responsibility for issuing this declaration of conformity.

Description of the device:
PAN-C v2

and its identical or similar derivatives.

Conformity with other directives / regulations / standards also applicable to the product is declared:

- RoHS-Richtlinie 2011/65/EU vom 08. Juni 2011
- EN IEC 62368-1:2020 + A11:2020
- EN 55032:2015 + AC:2016 + A11:2020 + A1:2020

Place and date of the exhibition: Hürth, 1 June 2024